

Problem C. Cut Tree

Time Limit: 10 seconds | Memory Limit: 512 MB

Standard Input / Standard Output

You are given a tree with n vertices. Vertex i has a weight z_i . The $n - 1$ edges are numbered from 1 to $n - 1$ in the order they are given in the input.

You have to answer q independent queries. A query is given by three integers l, r, f with $1 \leq l \leq f \leq r \leq n - 1$. For this query, build a graph on the n vertices that contains exactly the edges whose index lies in $[l, r]$, **except** that edge number f is deleted. Formally, the edge set is $\{i : l \leq i \leq r, i \neq f\}$.

This graph is a forest. The weight of a connected component is the sum of the weights of the vertices it contains (an isolated vertex forms a component of weight equal to its own weight). For each query output the maximum component weight.

Input

The first line contains one integer T ($1 \leq T \leq 10^5$), denoting the number of test cases. The description of the test cases follows.

The first line of each test case contains one integer n ($2 \leq n \leq 2 \times 10^5$).

Each of the next $n - 1$ lines contains two integers u_i and v_i ($1 \leq u_i, v_i \leq n, u_i \neq v_i$), denoting the two endpoints of edge i .

The next line contains n integers $z_1, z_2, \dots, z_n$ ($1 \leq z_i \leq 100$), denoting the vertex weights.

The next line contains one integer q ($1 \leq q \leq 2 \times 10^5$), denoting the number of queries.

Each of the next q lines contains three integers l, r, f ($1 \leq l \leq f \leq r \leq n - 1$) describing one query.

It is guaranteed that, within a single test file, the sum of n over all test cases does not exceed 2×10^5 , and the sum of q over all test cases does not exceed 2×10^5 .

Output

For each test case print q lines. The i -th of them must contain a single integer, denoting the answer to the i -th query of that test case.

Examples

Sample Input 1	Sample Output 1
2 5 1 2 2 3 3 4 4 5 10 20 30 40 50 3 1 4 2 2 4 4 1 1 1 2 1 2 7 5 1 1 1 1	120 90 50 7

Note

In the first test case the tree is a path $1 - 2 - 3 - 4 - 5$; edge i connects i and $i + 1$, and the weights are 10, 20, 30, 40, 50.

- Query (1, 4, 2): keep edges $\{1, 3, 4\}$. The component $\{1, 2\}$ has weight 30, and the component $\{3, 4, 5\}$ has weight 120; therefore, the maximum component weight is 120.
- Query (2, 4, 4): keep edges $\{2, 3\}$. The components $\{2, 3, 4\}$, $\{1\}$, and $\{5\}$ have weights 90, 10, and 50, respectively; therefore, the maximum component weight is 90.
- Query (1, 1, 1): no edge is kept, every vertex is isolated; the answer is the largest weight 50.

In the second test case the only edge is deleted by the query, so both vertices are isolated and the answer is $\max(7, 5) = 7$.